



Crypto Crime Network Analysis

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Abstract

- This project conducts two experiments utilizing separate cryptocurrency datasets to evaluate machine learning approaches for detecting cyber-enabled financial crime.
- These experiments will provide a focused assessment of how graph-based and transaction level features signal illicit activity across distinct crypto crime datasets. In addition, it will provide metrics of how consistent these indicators remain in different cybercriminal contexts.
- Elliptic Dataset: contains labeled licit and illicit Bitcoin transactions.
- Mendeley Dataset: contains scam-related transactions from non-bitcoin environments.



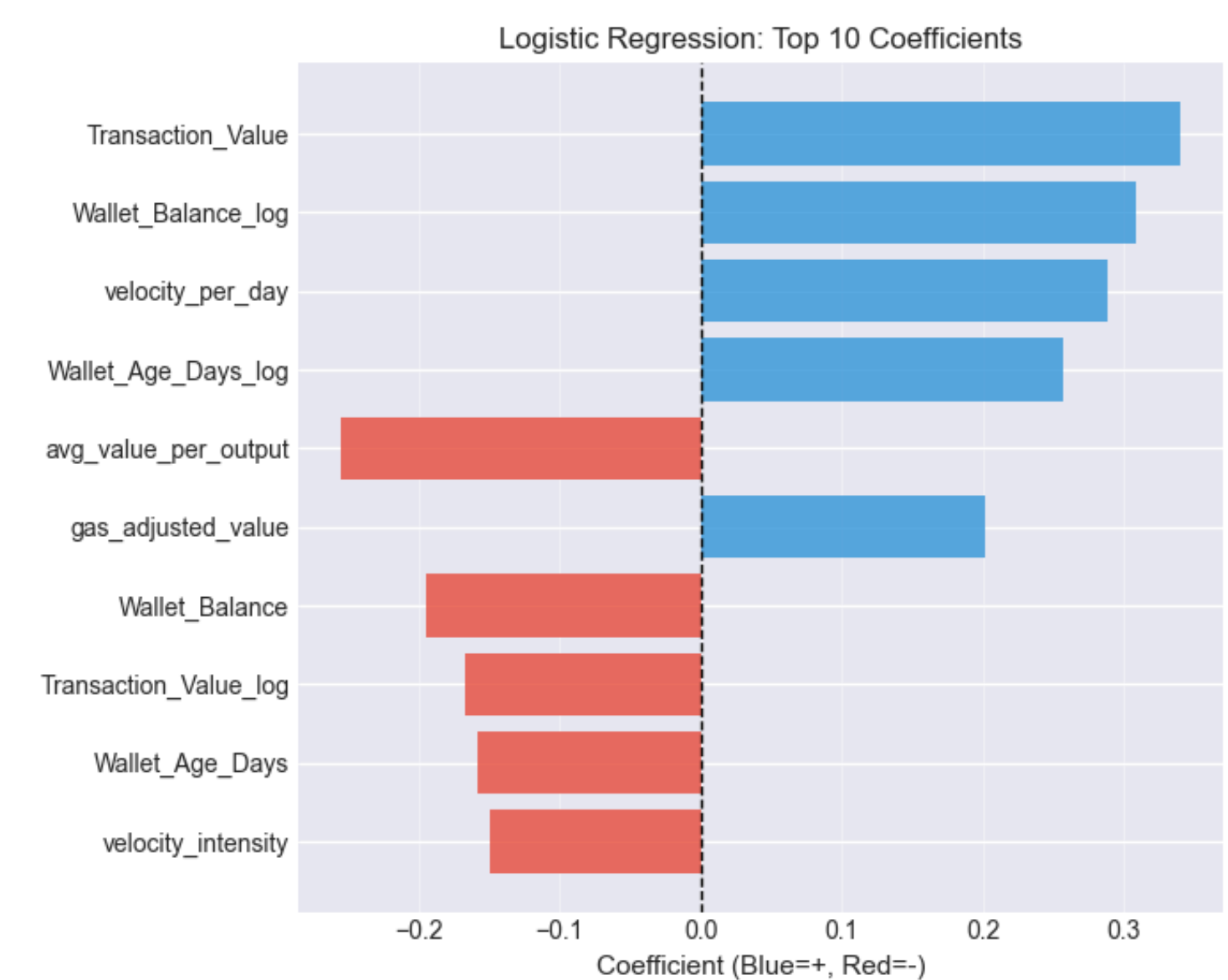
Research Questions:

- To what extent can graph-structural properties of Bitcoin transactions distinguish illicit activity from licit activity in a cryptocurrency transaction network?
- Do illicit transaction patterns found in the Elliptic dataset generalize to another form of cryptocurrency crime, such as scams?

Data Analysis



Illicit nodes (~2.0) have significantly lower average degree compared to **licit nodes (~3.1)**. This indicates that illicit transactions occur in sparser regions of the graph, suggesting its actors are potentially limiting their connections to avoid detection.



Transaction Value, Wallet Balance, and velocity per day have the largest coefficients. Despite these associations, overall model performance remains weak (F1 ~0.50), indicating that these behavioral patterns have limited discriminatory power for scam detection.

Methods

Datasets

Elliptic Dataset

- Contains 200K+ of only Bitcoin transactions
- Columns utilized:
 - Transaction features (30 cols)
 - Graph/behavioral/temporal nodes

Mendeley Dataset

- Contains 1,245 scam-related transactions other than Bitcoin currency
- Columns utilized:
 - Wallet activity
 - Behavioral ratios/intensity

Models

Random Forest Classifier/Regression

An ensemble model combining several decision trees by averaging (regression) or voting (classification).

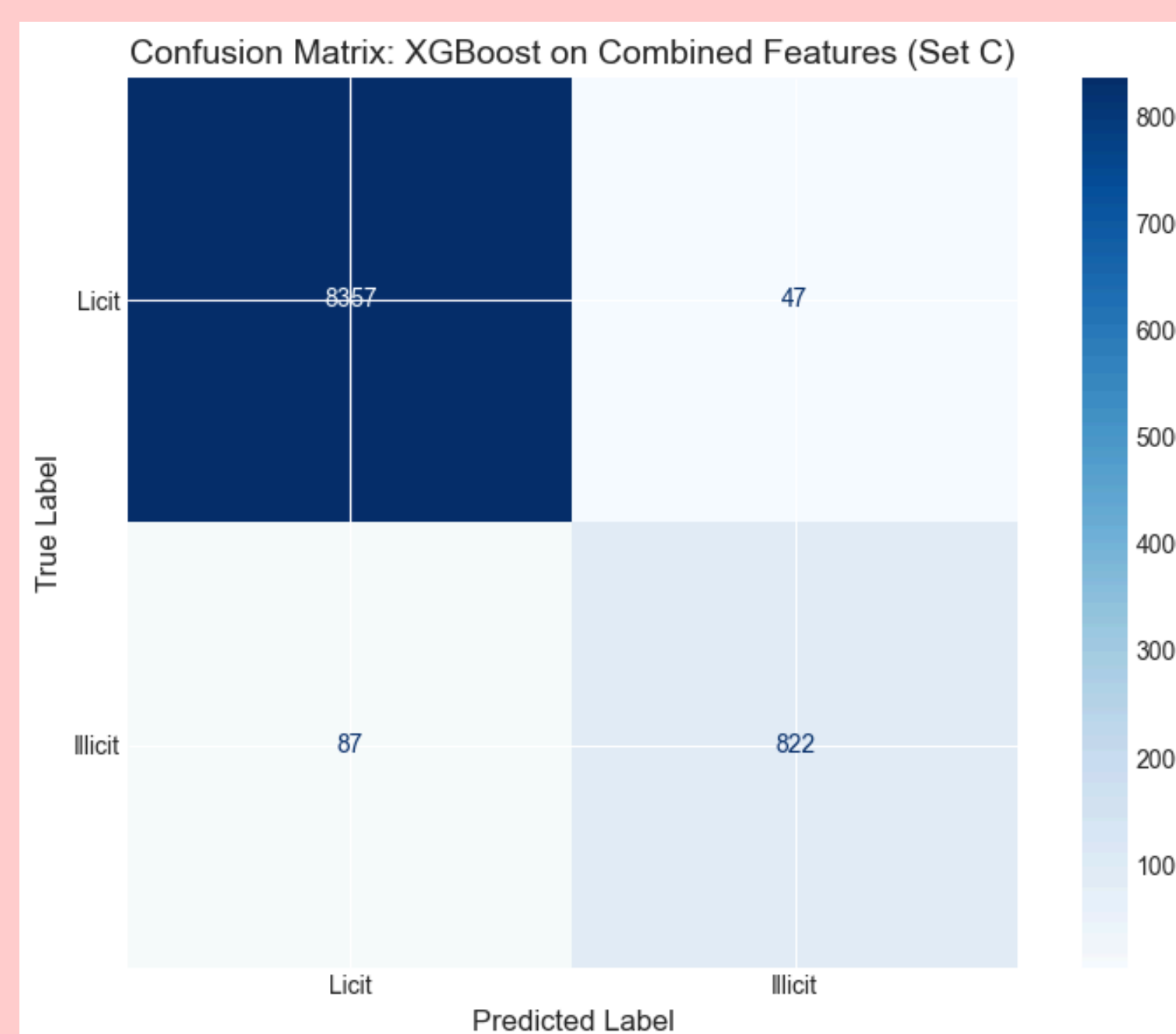
Logistic Regression

A simple, interpretable model that estimates the probability of a binary outcome by fitting a logistic function to the input features.

RQ1: Graph-structural, behavioral, and temporal features provide modest but complementary information that improves performance when combined with transaction features.

RQ2: Patterns show weak behavioral overlap rather than strong predictive generalization across crime types

Elliptic Model Results



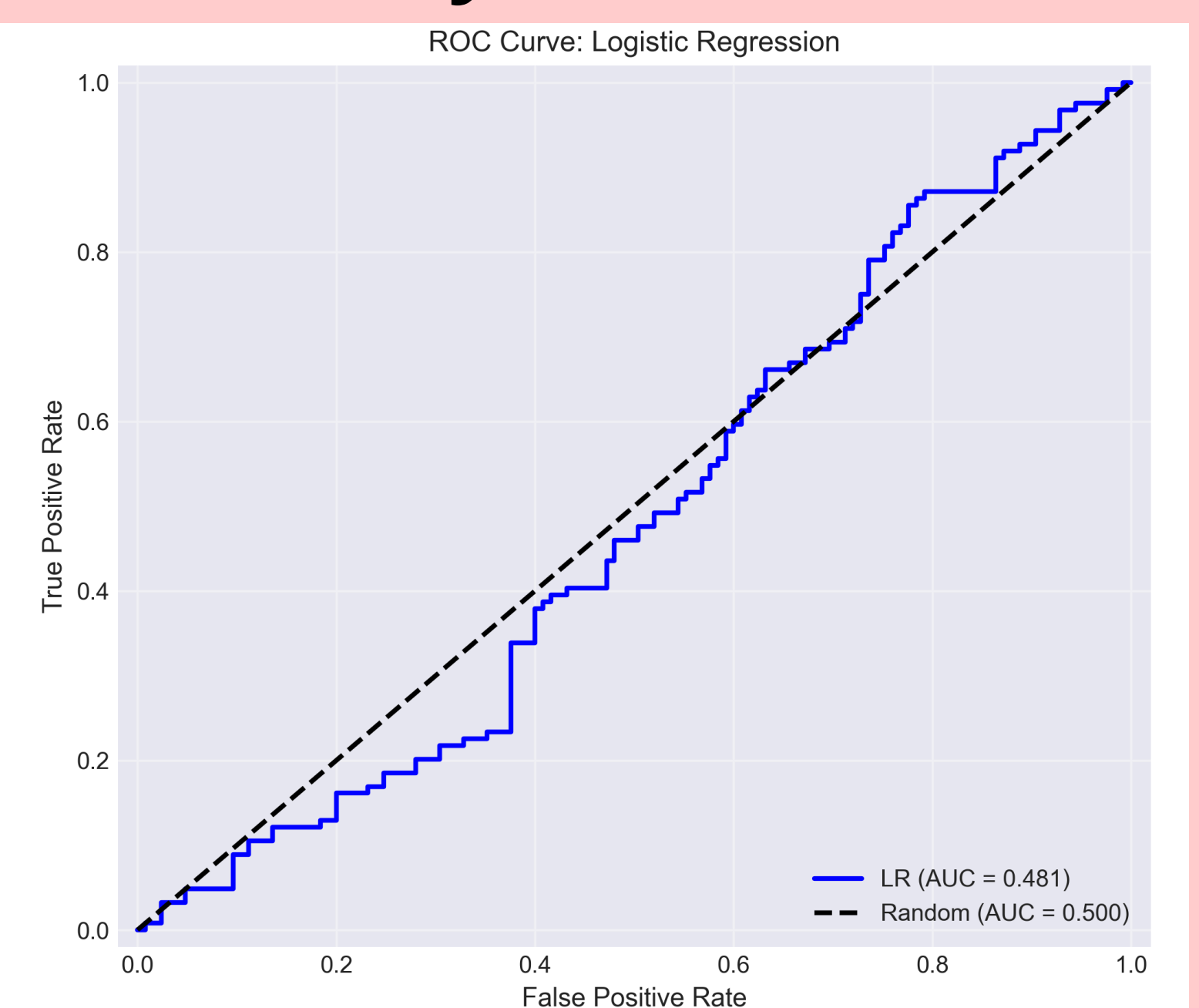
ROC-AUC: 0.99

Precision: 0.94

F1 Score: 0.92

Recall: 0.90

Mendeley Model Results



ROC-AUC: 0.48

Precision: 0.48

F1 Score: 0.47

Recall: 0.47

Code Base



What We Learned

- Elliptic's pre-computed transaction features carry the strongest signal for illicit detection
- Graph-structural features provide marginal improvement when combined with transaction features
- Ensemble methods (XGBoost with SMOTE) improve reliability on imbalanced data
- Illicit nodes tend to be in sparser graph regions with lower connectivity and clustering